PROJECT OVERVIEW

SkillPass BD

Digital professional identity & verified work history for Bangladesh's informal skilled workers

Features & tech stack at a glance · Based on PRD v1.2 · 25 October 2026

Users

Customer
Finds, books and reviews workers

Worker
Builds a "skill passport", takes and completes jobs

Admin
Verifies workers, resolves disputes, manages the platform

Contractor
Manages a team of workers
STRETCH

Core flow: Customer requests service → system matches workers → customer books → worker accepts → On the way → Start → Complete → customer confirms → review → **verified job added to worker's history + trust score updated**

MVP Features P0 — MUST SHIP

1. Accounts & Access
- Register with BD phone number
- Workers: 6-digit PIN · Customers: password
- JWT login, workers stay logged in 30 days
- Role-based access: customer / worker / admin
- Weak-PIN block, lockout, admin PIN reset

2. Worker SkillPass Profile
- Guided setup wizard with picture tiles
- Skills, experience, service areas
- One-tap availability switch
- Public profile with badges & trust score
- Profile completeness bar

3. Search & Matching
- Search by category, skill, area, trust, verified, available
- Rule-based matching: skill, trust, availability, experience, response time
- Each match shows why it matched

4. Requests & Bookings
- Customer service request (problem, area, time)
- Job lifecycle with status timeline
- Auto-expire (24 h) and auto-close (72 h)
- Cancel / dispute with reason
- Phone revealed only after acceptance

5. Work History & Reviews
- Customer-confirmed jobs become verified history
- Structured review: quality, professionalism, punctuality, communication
- "Would hire again" yes / no

6. Explainable Trust Score
- 0–100 score from 7 measurable factors
- Per-factor breakdown shown on profile
- Fair for new workers ("New on SkillPass")

7. Verification
- Identity and skill-evidence uploads (camera)
- Admin approves / rejects with reason
- Separate badges: identity ≠ skill ≠ verified jobs
- Private, access-controlled documents

8. Reports, Disputes & Notifications
- Report problems, profiles, reviews
- Admin resolves disputes, suspends users, audit log
- In-app notifications with unread badge
- New-job sound + vibration for workers

9. Worker-Friendly UI
- Bengali first, everyday words
- Picture tiles, big buttons, almost no typing
- One big button per job step
- Call customer + Google Maps directions
- Help screen: call / WhatsApp support
- Fast on budget Android phones

10. Admin Panel
- Dashboard with key stats & chart
- Verification queue + document viewer
- Disputes queue, users management
- Categories, skills, areas management

After MVP & Out of Scope

Next P1

- Bengali read-aloud (🔊)
- Voice typing
- Installable app (PWA)
- Bengali tutorial video
- Assisted onboarding by admin
- Booking chat messages
- Self-reported past work
- Weekly availability slots
- Email password reset
- Review replies, job photos
- Trust score history chart

Stretch P2

- AI: Bengali problem → service category
- AI: worker profile assistant
- Contractor / business module
- Redis + Celery background jobs
- SMS / push notifications

Not Included OUT

- Native Android / iOS apps
- Government NID verification
- Online payments
- Live GPS tracking
- Commercial-scale deployment

Tech Stack

Layer	Technology
Frontend	React 19 + Vite (JavaScript / JSX), React Router 7
Styling	Tailwind CSS 4, Bengali font (Hind Siliguri / Noto Sans Bengali)
Data & forms	TanStack Query, Axios (JWT refresh), React Hook Form + Zod
Language	react-i18next — Bengali / English
Charts	Recharts (admin dashboard)
Worker assistance	Web Speech API (read-aloud, voice typing), Vibration API, vite-plugin-pwa
Backend	Python 3.12, Django 5.2 LTS, Django REST Framework
Authentication	djangorestframework-simplejwt (rotation + blacklist)
Database	PostgreSQL 16+ (developed on 18)
API docs	drf-spectacular (OpenAPI + Swagger UI)
Other backend	django-filter, django-cors-headers, Pillow, django-environ
Testing	pytest, pytest-django, factory_boy · Vitest + React Testing Library
Code quality	ruff, black · ESLint, Prettier · pre-commit
Version control & CI	Git + GitHub, GitHub Actions
Local development	Local PostgreSQL · frontend can run alone on mock data
Deployment	Backend + DB: Render / Railway · Frontend: Vercel / Netlify · Media: cloud storage
Optional	Redis + Celery · LLM API for AI features

Team & delivery: 5 members. Lead builds database, backend, shared frontend and deployment. Members M2–M5 own frontend areas: Public & Auth (M2), Customer (M3), Worker (M4), Admin (M5). Built in 12 parts: Foundation → Auth → Catalog → Profiles & Search → Bookings → Reviews & Trust → Verification → Reports → Notifications & Help → Admin → Worker assistance → Final & Deploy.